A SPECIAL Π^0_1 CLASS WITH THE JOIN PROPERTY BUT WITHOUT PSEUDOJUMP INVERSION

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Abstract. Jananthan and Simpson showed that special Π^0_1 classes exist in three of the four possible cases determined by whether the Join Property and the Pseudojump Inversion Property hold, leaving open the existence of a special Π^0_1 class with the Join Property but not the Pseudojump Inversion Property. We construct such a class, thereby answering their question affirmatively and completing the four-way existence pattern.

Moreover, the class can be chosen to consist entirely of generalized low reals, with the Join Property holding locally on every nonempty relatively basic open subclass. More generally, every special Π^0_1 class contained in GL_1 has a special Π^0_1 extension, still contained in GL_1, on which the Join Property holds locally.

We also show that no class contained in GL_1 has the Pseudojump Inversion Property. An independent appendix proves that, for any class with the Join Property, every jump-inversion fibre above $0'$ is countably infinite.

1. Introduction and statements

At the level of Turing degrees, the Turing jump induces a map whose range is exactly the upper cone above $0'$. One inclusion is immediate, since $0' \leq_{\mathrm{T}} X'$ for every real X, while the converse is the content of Friedberg's Jump Inversion Theorem: every Turing degree above $0'$ has a preimage under the jump. A remarkable refinement, due to Jockusch and Soare [3, Theorem 1.4], says that, for every real $A \geq_{\mathrm{T}} 0'$ and every prescribed special Π^0_1 class $\mathcal{C}$, some $B \in \mathcal{C}$ satisfies $B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} A$.

The Posner–Robinson Join Theorem and the Jockusch–Shore Pseudojump Inversion Theorem are two closely related classical results concerning the interaction between the Turing jump, joining, and relative computable enumerability. This naturally raises the question whether analogous class-restricted forms of these two theorems hold.

Pseudojump operators were introduced and studied by Jockusch and Shore as generalizations of the Turing jump. In the form relevant here, they are the operators

$$J_e(X) = X \oplus W_e^X,$$

where W_e^X is the eth set computably enumerable in X. Their Pseudojump Inversion Theorem shows, in particular, that for each e there is a noncomputable c.e. set B such that

$$J_e(B) \equiv_{\mathrm{T}} 0'.$$

The usefulness of this generalization goes beyond the formal analogy with the jump: pseudojump inversion provides a flexible method for constructing Turing degrees with prescribed degree-theoretic properties.

Jockusch and Shore used this method to obtain, among other applications, finite-injury constructions at the various levels of the high and low hierarchies [4]. Their subsequent work developed finite and transfinite iterations of these operators through the REA hierarchy, together with general completeness and join theorems [5, Theorems 2.6 and 3.2].

A recurring question in the later theory has been which additional degree-theoretic constraints can be imposed on an inversion. Coles, Downey, Jockusch, and LaForte studied which extra properties can be imposed on sets completing a pseudojump operator and showed, in

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particular, that pseudojump completion by c.e. sets is not always compatible with upper cone avoidance [1]. Further work has continued this theme of compatibility between pseudojump inversion and additional requirements; see, for example, [2]. The problem considered here is a class-restricted version of the same general phenomenon: rather than imposing only a degree-theoretic side condition on the witness, we require the witness to lie in a prescribed special Π^0_1 class.

For standard background on classical computability theory and the c.e. degrees, see Soare [8]. For jump classes, jump inversion, and pseudojump operators (called *hops* there), see Odifreddi [6, §§XI.1, XIII.4].

We now fix notation. We work in Cantor space $2^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$, where $\mathbb{N} = \{0,1,2,\ldots\}$, and identify each real X with the subset $\{n : X(n) = 1\}$ of $\mathbb{N}$.

For reals X, Y, we write $X \leq_{\mathrm{T}} Y$, equivalently $Y \geq_{\mathrm{T}} X$, if membership in X is decidable by a Turing machine with oracle Y. We use the corresponding strict and equivalence relations:

$$\begin{aligned} X \equiv_{\mathrm{T}} Y &\iff X \leq_{\mathrm{T}} Y \text{ and } Y \leq_{\mathrm{T}} X, \\ X <_{\mathrm{T}} Y &\iff X \leq_{\mathrm{T}} Y \text{ and } Y \not\leq_{\mathrm{T}} X, \\ X >_{\mathrm{T}} Y &\iff Y <_{\mathrm{T}} X. \end{aligned}$$

In degree comparisons, 0 denotes the computable Turing degree, that is, the degree of $\varnothing$; thus

$$X >_{\mathrm{T}} 0 \iff 0 <_{\mathrm{T}} X \iff X \text{ is noncomputable.}$$

The *join* $X \oplus Y$ is obtained by interleaving the two reals:

$$(X \oplus Y)(2n) = X(n), \qquad (X \oplus Y)(2n+1) = Y(n).$$

An oracle for $X \oplus Y$ therefore gives access to both X and Y. Fix a standard effective enumeration $(\Phi_e)_{e\in\mathbb{N}}$ of oracle Turing machines. The *Turing jump* of X is

$$X' = \{e : \Phi_e^X(e)\downarrow\},$$

where $\downarrow$ denotes convergence. We write $0' = \varnothing'$ for the ordinary halting set.

A set is *computably enumerable* (c.e.) in X if it can be enumerated by a Turing machine with oracle X. Fix a standard effective enumeration $(W_e^X)_{e\in\mathbb{N}}$ of these sets, and write $W_e = W_e^{\varnothing}$ in the unrelativised case.

Let $2^{<\mathbb{N}}$ denote the set of finite binary strings. For $\sigma \in 2^{<\mathbb{N}}$ and $X \in 2^{\mathbb{N}}$, the notation $\sigma \prec X$ means that σ is an initial segment of X. The corresponding *cylinder* is

$$[\sigma] = \{X \in 2^{\mathbb{N}} : \sigma \prec X\}.$$

These cylinders form a basis for the topology of Cantor space. A *c.e. open subset of Cantor space* is a union of a computably enumerable family of cylinders. A Π^0_1 class is the complement in Cantor space of such an open set, and is *special* if it is nonempty and has no computable member. We write

$$\mathrm{GL}_1 = \{X \in 2^{\mathbb{N}} : X' \equiv_{\mathrm{T}} X \oplus 0'\}$$

for the class of *generalized low*, or GL_1, reals. The subscript 1 refers to the first Turing jump.

Jananthan and Simpson formulate the following class-restricted versions of the Join Theorem and the Pseudojump Inversion Theorem.

Definition 1.1 (Jananthan–Simpson [3])**.** A class $\mathcal{P} \subseteq 2^{\mathbb{N}}$ has the *Join Property*, denoted $\mathsf{JP}(\mathcal{P})$, if, whenever $A \geq_{\mathrm{T}} Z \oplus 0'$ and $Z >_{\mathrm{T}} 0$, some $B \in \mathcal{P}$ satisfies

$$A \equiv_{\mathrm{T}} B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} B \oplus Z. \tag{1.1}$$

It has the *Pseudojump Inversion Property*, denoted $\mathsf{PJI}(\mathcal{P})$, if, for every $A \geq_{\mathrm{T}} 0'$ and for all $e \in \mathbb{N}$, some $B \in \mathcal{P}$ satisfies

$$A \equiv_{\mathrm{T}} J_e(B) \equiv_{\mathrm{T}} B \oplus 0'. \tag{1.2}$$

Thus the Join Property requires the witness in the Posner–Robinson Join Theorem [7] to lie in a prescribed class, while the Pseudojump Inversion Property imposes the corresponding restriction on the Jockusch–Shore Pseudojump Inversion Theorem [4].

Jananthan and Simpson realized three of the four logically possible combinations of these two properties for special Π^0_1 classes. The class CPA of complete consistent extensions of Peano Arithmetic has both properties. There are special Π^0_1 classes with the Pseudojump Inversion Property but without the Join Property, and they construct a special Π^0_1 class having neither property. The remaining possibility was left open:

	PJI	$\neg\mathsf{PJI}$
JP	CPA	open in [3]
$\neg\mathsf{JP}$	known	known.

Equivalently, they ask whether there exists a special Π^0_1 subclass of Cantor space which has the Join Property but not the Pseudojump Inversion Property [3, Introduction].

Our main theorem supplies precisely this missing case and therefore completes the four-way existence pattern for the two properties.

Theorem 1.2. *There is a special Π^0_1 class $\mathcal{P} \subseteq 2^{\mathbb{N}}$ such that*

$$\mathcal{P} \subseteq \mathrm{GL}_1, \qquad \mathsf{JP}(\mathcal{P}), \qquad \neg\mathsf{PJI}(\mathcal{P}).$$

More strongly, the Join Property holds locally on $\mathcal{P}$: for every finite binary string σ such that $\mathcal{P} \cap [\sigma] \neq \varnothing$, the class $\mathcal{P} \cap [\sigma]$ has the Join Property.

The generalized-lowness requirement plays two roles in the argument. First, the construction ensures that every member of the limit class $\mathcal{P}$ is generalized low. Second, generalized lowness provides the separation: no class contained in GL_1 can have the Pseudojump Inversion Property. Thus the main work is to construct a special Π^0_1 class with the Join Property whose members all belong to GL_1.

In fact, the construction gives a stronger extension theorem.

Theorem 1.3 (Extension theorem)**.** *Let $\mathcal{Q} \subseteq 2^{\mathbb{N}}$ be a special Π^0_1 class with $\mathcal{Q} \subseteq \mathrm{GL}_1$. There is a special Π^0_1 class $\mathcal{P}$ such that*

$$\mathcal{Q} \subseteq \mathcal{P} \subseteq \mathrm{GL}_1,$$

and, for every $\sigma \in 2^{<\mathbb{N}}$ such that $\mathcal{P} \cap [\sigma] \neq \varnothing$, the class $\mathcal{P} \cap [\sigma]$ has the Join Property.

A class $\mathcal{Q}$ satisfying the hypothesis exists by [3, Theorem 6.2, parts 2 and 4]. Only generalized lowness of the individual members is needed here, not the stronger assertion there concerning finite joins.

The construction uses a computable decreasing sequence of clopen classes. Copies of the initial class $\mathcal{Q}$ are retained at the limiting points of an infinitely branching system of choices. A finite-injury mechanism eventually decides, at each address and exit, whether the relevant c.e. open requirement is met; a permanent historical record of the actions, indexed by epochs, later permits the required decoding. The two decoding procedures are arranged so that the same constructed path B satisfies

$$A \equiv_{\mathrm{T}} B \oplus Z \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} B'.$$

The construction is local in the sense expressed in Theorem 1.2: every nonempty relatively basic open part of $\mathcal{P}$ again has the Join Property.

We prove the extension theorem in Sections 2–6. Section 5 establishes generalized lowness and specialness, and Section 7 derives the failure of pseudojump inversion and completes the proof of the main theorem. The appendix is independent of the construction: it shows that, for any class with the Join Property, every jump-inversion fibre above $0'$ represents countably infinitely many Turing degrees.

2. Exits, addresses, and requirements

For a finite sequence σ, its length is denoted by $|\sigma|$, and $\langle\rangle$ denotes the empty sequence. All coordinates are indexed starting at 0. For finite sequences over the same alphabet, $\sigma \preceq \tau$ means that σ is a prefix of τ, while $\sigma \prec \tau$ means that it is a proper prefix. We use $\succeq$ for the reverse prefix relation. Two finite sequences are *incompatible* if neither is a prefix of the other. For a finite or infinite sequence α, the notation $\alpha \upharpoonright n$ denotes its prefix of length n, whenever defined.

The operation $^\frown$ denotes concatenation. For binary strings, $\sigma ^\frown b$ appends the single bit b, and $\sigma ^\frown X$ prefixes the real X by σ. More generally, for a class $\mathcal{C} \subseteq 2^{\mathbb{N}}$, put

$$\sigma ^\frown \mathcal{C} = \{\sigma ^\frown X : X \in \mathcal{C}\}.$$

Thus $\sigma ^\frown \mathcal{C}$ is a prefixed copy of $\mathcal{C}$ inside $[\sigma]$. We write 0^ω for the constant zero real.

A set is *clopen* if it is both open and closed; in Cantor space, these are exactly the finite unions of cylinders. A binary tree is a subset of $2^{<\mathbb{N}}$ closed under prefixes. For such a tree T, let

$$[T] = \{X \in 2^{\mathbb{N}} : (\forall n)\ X \upharpoonright n \in T\}$$

be its class of infinite paths.

Fix $\mathcal{Q}$ as in Theorem 1.3. Choose a computable tree $T \subseteq 2^{<\mathbb{N}}$ with $\mathcal{Q} = [T]$ and $\langle\rangle \in T$. We do not assume that T is pruned: some nodes of T may lie on no infinite path.

For a nonempty string η, write $\eta^- = \eta \upharpoonright (|\eta| - 1)$ for its immediate predecessor. Let

$$E = \{\eta \in 2^{<\mathbb{N}} \setminus T : \eta^- \in T\}. \tag{2.1}$$

The members of E are called *exits* from T. A set of strings is *prefix-free* if no two distinct members are comparable under $\preceq$.

Lemma 2.1. *The set E is computable, infinite, and prefix-free. Every real either belongs to $\mathcal{Q}$ or extends exactly one member of E. If $\mathcal{Q} \cap [\delta] \neq \varnothing$, some member of E extends δ.*

Proof. Computability follows from that of T. A string in E is the first prefix outside T of any real extending it, which proves prefix-freeness and the asserted partition. If E were finite, then $\mathcal{Q} = 2^{\mathbb{N}} \setminus \bigcup_{\eta \in E} [\eta]$ would be a nonempty clopen class and would contain a computable real, a contradiction. Finally, if $\mathcal{Q} \cap [\delta] \neq \varnothing$, then $\delta \in T$. The computable real $\delta ^\frown 0^\omega$ does not belong to $\mathcal{Q}$. Its first prefix outside T therefore strictly extends δ and belongs to E. □

Fix a computable enumeration without repetitions $E = \{\eta_i : i \in \mathbb{N}\}$. Given an oracle for a real outside $\mathcal{Q}$, its exit index i can be recovered: find its shortest prefix outside T and search the enumeration for that prefix.

Our *addresses* are elements of

$$\mathcal{A} = (\mathbb{N} \times \{0,1\})^{<\mathbb{N}}.$$

The length $|a|$ of an address is the number of pairs occurring in it.

The same prefix conventions apply to addresses. We use fixed computable codings of binary strings, addresses, and finite tuples by natural numbers whenever these objects are used as inputs to algorithms. We write $a ^\frown (k, b)$ for the address obtained by appending the pair (k, b) to a, where $k \in \mathbb{N}$ and $b \in \{0, 1\}$. Here k records an exit index and b is a binary choice.

A *slot* is a pair (a, k), where $a \in \mathcal{A}$ and $k \in \mathbb{N}$. The two immediate child addresses associated with the slot (a, k) are

$$a ^\frown (k, 0) \qquad \text{and} \qquad a ^\frown (k, 1).$$

For example, if

$$a = \langle (4, 0), (7, 1) \rangle,$$

then the slot $(a, 3)$ has the two immediate child addresses

$$a ^\frown (3, 0) \qquad \text{and} \qquad a ^\frown (3, 1).$$

Thus the natural number 3 records the chosen exit, while the last coordinate 0 or 1 records the subsequent binary choice.

A slot (c, ℓ) is *below* (a, k) if its address c extends one of these two child addresses; equivalently,

$$a^\frown(k, b) \preceq c \quad \text{for some } b \in \{0, 1\}. \tag{2.2}$$

This relation is computable. In particular, the slot (a, k) itself is not below (a, k).

A slot (d, j) is a *proper ancestor slot* of an address a if a extends one of the two child addresses associated with (d, j); equivalently,

$$d^\frown(j, b) \preceq a \quad \text{for some } b \in \{0, 1\}.$$

Each address a will be represented in the ambient binary tree $2^{<\mathbb{N}}$ by a finite binary string, called its *physical root*. During the construction its value may change; its value at stage s will be denoted by $p_s(a)$. We shall prove that $p_s(a)$ eventually stabilises and denote its final value by $p(a)$. The binary-string length $|p_s(a)|$ (and, eventually, $|p(a)|$) should not be confused with the address length $|a|$. Its role is to locate, inside Cantor space, the prefixed copy of $\mathcal{Q}$ associated with the address a.

We use the following uniformly c.e. sequence $(U_n)_{n\in\mathbb{N}}$ of open sets, meaning that one algorithm, given n, enumerates cylinders whose union is U_n:

$$U_{2e} = \{X : \Phi_e^X(e)\downarrow\}, \tag{2.3}$$

$$U_{2e+1} = \{X : \exists m\,[\varphi_e(m)\downarrow\in\{0,1\}\ \&\ X(m) \neq \varphi_e(m)]\}. \tag{2.4}$$

Here (φ_e) is a standard effective enumeration of partial computable functions. The notation $\varphi_e(m)\downarrow\in\{0,1\}$ means that the computation converges with a binary output.

Fix uniformly computable increasing clopen approximations $U_{n,s}$ with $U_n = \bigcup_s U_{n,s}$. Concretely, a finite list of cylinders describing $U_{n,s}$ can be computed from n, s, and $U_{n,s} \subseteq U_{n,s+1}$. Slots at addresses of length n will decide U_n on their surviving paths. The even requirements control the jump; the odd requirements exclude computable complete paths.

3. The computable construction

The construction proceeds in stages. At each stage we keep a finite table recording the actions which are still current, together with a permanent record of all actions that have ever occurred. The current table determines the locations of the copies of $\mathcal{Q}$ inside Cantor space and hence the current clopen class $\mathcal{P}_s$. At each stage one slot is visited. If it acts, the construction restricts the corresponding exit region to a chosen cylinder and reinitialises all descendant slots.

3.1. The data at stage s.

At the beginning of stage $s+1$ we have a finite partial table D_s. Its domain $\mathrm{dom}(D_s)$ is the finite set of slots with a current stored entry. An entry

$$D_s(a, i) = (\rho, u)$$

records the stem ρ selected by an action at stage $u > 0$ which is still current. Every action has its own stage number as identifier; identifiers are never reused. An action at an ancestor deletes the current entries in the slots below it.

There is also a permanent historical record of events

$$(a, t, i, \rho, u). \tag{3.1}$$

This says that (a, i) acted at stage u, during epoch t of a, and selected ρ. Nothing is ever removed from this record. Its purpose is distinct from that of D_s: an obsolete action disappears from the current table but remains in the historical record. From the current table D_s we derive four kinds of data. The string $p_s(a)$ is the current physical root of the address a; $r_s(a, i)$ is the current exit root obtained by appending η_i to $p_s(a)$; $g_s(a, i)$ is the current end of the stem attached to that exit; and $\theta_s(a)$ records the current version, or epoch, of the address.

Formally, these data are defined from D_s by recursion on address length. Set

$$p_s(\langle\rangle) = \langle\rangle, \qquad \theta_s(\langle\rangle) = 0,$$

and put

$$r_s(a,i) = p_s(a) \frown \eta_i, \tag{3.2}$$

$$g_s(a,i) = \begin{cases} \rho, & D_s(a,i) = (\rho, u), \\ r_s(a,i), & (a,i) \notin \mathrm{dom}(D_s), \end{cases} \tag{3.3}$$

$$p_s(a \frown (i,b)) = g_s(a,i) \frown b, \tag{3.4}$$

$$\theta_s(a \frown (i,b)) = \begin{cases} u, & D_s(a,i) = (\rho, u), \\ \theta_s(a), & (a,i) \notin \mathrm{dom}(D_s). \end{cases} \tag{3.5}$$

Only the finitely many prefixes of a are consulted to evaluate these functions on a given input. We maintain that every current stem extends its current exit root. The values $p_s(a)$, $r_s(a,i)$, and $g_s(a,i)$ are finite binary strings, whereas $\theta_s(a) \in \mathbb{N}$ identifies the current epoch. This label need not equal the current stage s and does not measure the length of any of these strings.

The current clopen class is given by the finite table:

$$\mathcal{P}_s = 2^{\mathbb{N}} \setminus \bigcup_{(a,i) \in \mathrm{dom}(D_s)} ([r_s(a,i)] \setminus [g_s(a,i)]). \tag{3.6}$$

Under the maintained stem condition, a current record at (a,i) excludes from the exit cylinder $[r_s(a,i)]$ everything outside the stem cylinder $[g_s(a,i)]$; if the two strings are equal, this exclusion is empty.

A clopen set here is given by an explicit finite Boolean combination of cylinders, so emptiness and inclusion are decidable.

3.2. **Initialisation and scheduling.** Initially D_0 and the historical record are empty, and $\mathcal{P}_0 = 2^{\mathbb{N}}$. We use a simple fair schedule: exactly one slot is visited at each stage, and every slot is visited infinitely often. Fix a computable bijection ν from $\mathbb{N}$ to the slots. At stage $s+1$ visit the slot

$$\nu(v_2(s+1)),$$

where $v_2(m)$ is the exponent of 2 in the positive integer m. There is therefore at most one action per stage. Each individual stage terminates: all tests performed at that stage are decidable, and any search undertaken after a positive test is guaranteed to succeed. We impose no a priori running-time bound.

3.3. **The action.** An action has two effects: it selects a cylinder contained in the current exit region, and it reinitialises the descendant slots by deleting their current entries. The corresponding historical events are retained in the permanent record.

Suppose the visited slot is (a,i). If it is already in $\mathrm{dom}(D_s)$, do nothing. Otherwise put

$$n = |a|, \qquad t = \theta_s(a), \qquad r = r_s(a,i),$$

and test whether the clopen set

$$C = \mathcal{P}_s \cap [r] \cap U_{n,s} \tag{3.7}$$

is nonempty. If it is empty, again do nothing. If it is nonempty, choose, in length-lexicographic order, a string $\rho \succeq r$ with

$$[\rho] \subseteq C. \tag{3.8}$$

Such a string exists because C is nonempty and clopen. The search is computable, since inclusion of clopen sets is decidable.

Perform the following update. Delete from D_s all entries in slots below (a,i), retain all other entries, and insert

$$D_{s+1}(a,i) = (\rho, s+1).$$

Add $(a,t,i,\rho,s+1)$ to the permanent historical record. The intended update to the clopen class is

$$\mathcal{P}_{s+1} = (\mathcal{P}_s \setminus [r]) \cup [\rho]. \tag{3.9}$$

Lemma 4.1 below verifies that the new table describes exactly this class. If no action is performed, both the table and the clopen class remain unchanged.

Every action counts as an action even if $\rho = r$. In particular, it still receives a new identifier and reinitialises the descendants. The epoch is a version identifier, not a code for the physical string.

4. Invariants, finite injury, and decisions

The verification has four steps. First we check the stage invariants and the monotonicity of the clopen approximations. Next we prove finite injury and obtain final roots, stems, and epochs. We then describe the paths through the limit class. Finally, we show that each exit at a final epoch decides the corresponding c.e. open requirement.

4.1. **Stage geometry and invariants.**

Lemma 4.1 (Stage invariants)**.** *The construction is computable, and at every stage the following properties hold:*

(i) *every stored stem extends its current exit root;*
(ii) *the new table at an action gives precisely* (3.9)*, and hence* $\mathcal{P}_{s+1} \subseteq \mathcal{P}_s$*;*
(iii) *for every* $a \in \mathcal{A}$*,*

$$p_s(a) \frown \mathcal{Q} \subseteq \mathcal{P}_s; \tag{4.1}$$

(iv) *for every* $a \in \mathcal{A}$ *and* $i \in \mathbb{N}$*, the exit* i *at* a *has a surviving continuation:*

$$\varnothing \neq p_s(a \frown (i,0)) \frown \mathcal{Q} \subseteq \mathcal{P}_s \cap [r_s(a,i)]. \tag{4.2}$$

Proof. We argue by induction on the stages. Initially $D_0 = \varnothing$ and $\mathcal{P}_0 = 2^{\mathbb{N}}$, so the stem condition and the two assertions about copies of $\mathcal{Q}$ hold.

Geometric preliminaries. First observe some geometric consequences of the stem condition. For every address a, exit i, and bit $b < 2$,

$$p_s(a) \frown \eta_i = r_s(a,i) \preceq g_s(a,i) \prec p_s(a \frown (i,b)).$$

Consequently, if $a \frown (i,b) \preceq c$, then $g_s(a,i) \frown b \preceq p_s(c)$. The two children of a slot have incompatible roots. Distinct exits at the same address also have incompatible roots, because E is prefix-free. It follows, by considering the first pair at which two addresses differ, that incompatible addresses have disjoint root cylinders.

Effect of an action. Suppose an action is performed at (a,i) at stage $s+1$. By the action rule, $(a,i) \notin \mathrm{dom}(D_s)$. Put $r = r_s(a,i)$, and let $\rho \succeq r$ be the chosen string, so

$$[\rho] \subseteq \mathcal{P}_s \cap [r] \cap U_{|a|,s}.$$

Partition the old records into those retained and those deleted. The point is to show that deleting the descendant records does not restore any real which had already been removed from $\mathcal{P}_s$. All exclusions considered in this paragraph are evaluated using the old physical roots and stems.

A retained record is either at an ancestor slot traversed on the way to a, or lies outside the descendant region of (a,i). In the latter case its root is incompatible with the exit root r. Its root depends only on records along its own address, none of which is changed by this action. Thus its physical root and its stored stem remain unchanged. Its excluded set is disjoint from $[r]$: in the ancestor case, $[r]$ is contained in its retained stem cylinder; in the incompatible case, the two root cylinders are disjoint.

Every deleted record is below (a,i), so its old root cylinder, and hence its excluded set, is contained in $[r]$. Its excluded set is also disjoint from $[\rho]$, since $[\rho] \subseteq \mathcal{P}_s$. Therefore every deleted exclusion is contained in

$$H = [r] \setminus [\rho].$$

The new table satisfies the stem condition. Indeed, all retained records have unchanged roots and stems. The root of (a, i) itself is still r, since its root depends only on proper ancestors, and its new stem is $\rho \succeq r$. All old records below (a, i) have been deleted.

Let F be the union of the exclusions of the retained records, and let G be the union of the exclusions of the deleted records. Then

$$2^{\mathbb{N}} \setminus \mathcal{P}_s = F \cup G \qquad \text{and} \qquad G \subseteq H.$$

Thus every exclusion which disappears from the table is already absorbed by the new exclusion H. The new table has exactly the retained exclusions together with the new exclusion H. Hence it describes

$$\begin{aligned} 2^{\mathbb{N}} \setminus (F \cup H) &= 2^{\mathbb{N}} \setminus (F \cup G \cup H) \\ &= \mathcal{P}_s \setminus H \\ &= (\mathcal{P}_s \setminus [r]) \cup [\rho], \end{aligned}$$

where the last equality uses $[\rho] \subseteq \mathcal{P}_s$. This proves (3.9) and monotonicity. If no action is performed, the table and the class are unchanged.

Survival of the copies of $\mathcal{Q}$. We next verify the copy assertions for any stage t whose table satisfies the stem condition. Fix an address c and a stored slot (d, j).

If $d^\frown(j, b) \preceq c$ for some $b < 2$, then

$$[p_t(c)] \subseteq [g_t(d, j)],$$

so the exclusion at (d, j) does not meet $p_t(c)^\frown \mathcal{Q}$.

If $d = c$, its exclusion is contained in $[p_t(c)^\frown \eta_j]$, which is disjoint from $p_t(c)^\frown \mathcal{Q}$ because $\eta_j \notin T$ and $\mathcal{Q} = [T]$. If $c \prec d$, let (k, b) be the first pair of d after c. The exclusion at (d, j) is then contained in $[p_t(c)^\frown \eta_k]$, and is again disjoint from $p_t(c)^\frown \mathcal{Q}$.

In all remaining cases, the slot (d, j) lies in a region incompatible with the address c, so its root cylinder is disjoint from $[p_t(c)]$. Thus no exclusion in the table meets $p_t(c)^\frown \mathcal{Q}$. Formula (3.6) proves (4.1).

Taking $c = a^\frown(i, 0)$ gives (4.2), since

$$p_t(c) = g_t(a, i)^\frown 0 \succeq r_t(a, i)$$

and $\mathcal{Q}$ is nonempty.

Effectivity. Finally, all stage data are computable. The table is finite, roots and epochs are evaluated by recursion on finite addresses, and the relation specifying which records to delete is computable. The set tested at an action is an explicitly presented clopen set, so its emptiness and the inclusion of any proposed cylinder are decidable. If the test is positive, this clopen set contains a cylinder, so the length-lexicographic search for ρ terminates. The finite table update and the addition to the historical record are effective. This completes the induction. □

4.2. **Finite injury and final values.** The epochs serve as finite-injury version numbers: an address receives a new epoch exactly when a proper ancestor slot acts, thereby reinitialising it.

Lemma 4.2 (Epochs and stabilisation)**.** *For each address a, the epoch $\theta_s(a)$ changes only when a proper ancestor slot acts, and then changes to that action's new identifier. It never returns to an earlier value. Each slot acts at most once in each epoch of its address. Every address has only finitely many epochs. In particular, the limits*

$$p(a) = \lim_s p_s(a), \qquad \theta(a) = \lim_s \theta_s(a), \qquad g(a, i) = \lim_s g_s(a, i) \tag{4.3}$$

exist for all a, i.

Proof. Changes of epoch. When (a, i) acts at stage u, every table entry below it is deleted. The epoch recursion therefore gives epoch u to all its descendant addresses. This identifier exceeds all earlier identifiers. Epochs outside this descendant region are unchanged. In particular an action at (a, i) does not change the epoch of a itself. Physical roots can change only in the

same descendant region. Thus a root is constant throughout a given epoch of its address, even though an action need not change its physical string.

Finite injury. Once a slot has acted, its entry remains until a proper ancestor acts. Such an action changes the epoch of its address. Hence the slot acts at most once per epoch.

The root address has a single epoch, namely 0, so each of its slots acts at most once. Inductively, if a has finitely many epochs, then a fixed slot (a, i) acts finitely often. Its actions, together with the finitely many changes of the epoch of a, account for all reinitialisations of each child $a^\frown(i, b)$. The child therefore has finitely many epochs as well. This completes the induction on address length.

Final values. For each fixed address a, its epoch is eventually constant. Since its physical root is constant throughout each epoch, $p_s(a)$ is eventually constant as well. Fix also an exit i. During the final epoch of a, the exit root $r_s(a, i) = p_s(a)^\frown\eta_i$ is fixed, and the slot (a, i) acts at most once. If it acts, its new record can never subsequently be deleted, since such a deletion would require a further change of the epoch of a. Thus $g_s(a, i)$ is eventually constant. This proves all the limits in (4.3).

The stabilisation is pointwise: no simultaneous stabilisation of all addresses or slots at a level is asserted.

□

The notation without a stage subscript, namely $p(a)$, $\theta(a)$, and $g(a, i)$, will henceforth denote these final values. The limits in (4.3) mean eventual constancy at each fixed input; they are not unions of the stage approximations.

4.3. **The limit class and its paths.** We now pass to the limit and define

$$\mathcal{P} = \bigcap_s \mathcal{P}_s. \tag{4.4}$$

It is a Π^0_1 class, since the complements of the uniformly computable clopen sets $\mathcal{P}_s$ enumerate an effectively open set. We first verify that every final copy of $\mathcal{Q}$ survives, and then use these copies to describe all paths through $\mathcal{P}$.

Lemma 4.3 (Surviving copies and paths)**.** *For every address a,*

$$p(a)^\frown\mathcal{Q} \subseteq \mathcal{P}. \tag{4.5}$$

In particular $\mathcal{Q} \subseteq \mathcal{P}$ and $\mathcal{P}$ is nonempty. Every $B \in \mathcal{P}$ has exactly one of the following two descriptions under the parsing rule below:

(i) *the procedure stops at an address a, and $B = p(a)^\frown R$ for some $R \in \mathcal{Q}$;*
(ii) *the procedure continues at every level, giving addresses $a_0 = \langle\rangle$ and $a_{n+1} = a_n^\frown(i_n, b_n)$ with $p(a_n) \prec B$ for all n.*

We call these paths stopped *and* complete*, respectively. This is a structural description; decidability of membership in $\mathcal{Q}$ is not assumed. Conversely, any infinite chain of addresses of the form in* (ii) *gives a member of $\mathcal{P}$ by taking the union of its final roots.*

Proof. Survival of the final copies. Fix an address a. By Lemma 4.2, there is a stage s_a such that $p_s(a) = p(a)$ for every $s \geq s_a$. Then (4.1) gives

$$p(a)^\frown\mathcal{Q} \subseteq \mathcal{P}_s \qquad (s \geq s_a).$$

For $s < s_a$, monotonicity gives $\mathcal{P}_{s_a} \subseteq \mathcal{P}_s$, so the same inclusion holds. Hence

$$p(a)^\frown\mathcal{Q} \subseteq \bigcap_s \mathcal{P}_s = \mathcal{P},$$

proving (4.5). Since $p(\langle\rangle) = \langle\rangle$ and $\mathcal{Q}$ is nonempty, we also have $\mathcal{Q} \subseteq \mathcal{P}$ and $\mathcal{P} \neq \varnothing$.

Parsing a member of $\mathcal{P}$. By the stage recursions, the stem condition, and pointwise stabilisation, for every a, i, and $b < 2$,

$$p(a)^\frown\eta_i \preceq g(a, i), \qquad p(a^\frown(i, b)) = g(a, i)^\frown b.$$

We also claim that

$$\mathcal{P} \cap [p(a) \frown \eta_i] \subseteq [g(a,i)] \qquad \text{for every } a, i. \tag{4.6}$$

To see this, choose s sufficiently large that $p_s(a) = p(a)$ and $g_s(a,i) = g(a,i)$. If $(a,i) \notin \mathrm{dom}(D_s)$, then $g(a,i) = r_s(a,i) = p(a) \frown \eta_i$, and the claim is immediate. Otherwise the table excludes

$$[r_s(a,i)] \setminus [g_s(a,i)]$$

from $\mathcal{P}_s$. Since $\mathcal{P} \subseteq \mathcal{P}_s$, (4.6) follows in this case as well.

Now let $B \in \mathcal{P}$ and start the parsing at $\langle\rangle$. Suppose it has reached a with $p(a) \prec B$, and write $B = p(a) \frown R_a$. If $R_a \in \mathcal{Q}$, the parsing stops. Otherwise Lemma 2.1 gives a unique i such that $\eta_i \prec R_a$. By (4.6), B extends $g(a,i)$. The bit

$$b = B(|g(a,i)|)$$

is therefore defined, and

$$p(a \frown (i,b)) = g(a,i) \frown b \prec B.$$

Thus the next address is uniquely determined. This establishes exhaustiveness and uniqueness of the parsing description. The parsing is understood structurally; no decision procedure for membership in $\mathcal{Q}$ is being asserted.

From an infinite address chain to a path. Conversely, let $a_0 = \langle\rangle$ and $a_{n+1} = a_n \frown (i_n, b_n)$ be any infinite address chain. The final-root identities give

$$p(a_n) \frown \eta_{i_n} \preceq g(a_n, i_n) \prec p(a_{n+1}).$$

Since every exit η_i is nonempty,

$$|p(a_{n+1})| \geq |p(a_n)| + |\eta_{i_n}| + 1 \geq |p(a_n)| + 2.$$

The roots therefore have an infinite union

$$B = \bigcup_n p(a_n).$$

Fix $R \in \mathcal{Q}$. By (4.5), every $p(a_n) \frown R$ belongs to $\mathcal{P}$. These reals converge to B, because they agree with B on the increasingly long prefixes $p(a_n)$. The class $\mathcal{P}$ is closed, so $B \in \mathcal{P}$.

For every n, the tail of B after $p(a_n)$ extends η_{i_n} and hence does not belong to $\mathcal{Q}$. Moreover, the bit of B immediately after $g(a_n, i_n)$ is b_n. Its unique parsing therefore follows the specified chain and never stops. □

4.4. **Event sets and final decisions.** For a fixed address and epoch, we now record which exits actually acted during that epoch.

Definition 4.4 (The event sets)**.** For $a \in \mathcal{A}$ and $t \in \mathbb{N}$, let

$$S_{a,t} = \{i : \exists \rho, u \ (a, t, i, \rho, u) \text{ occurs in the historical record}\}. \tag{4.7}$$

The family $(S_{a,t})_{a,t}$ is uniformly c.e., including when t is not an epoch actually attained by a. By the *s-m-n* theorem, fix a computable function q such that $S_{a,t} = W_{q(a,t)}$ for all a, t. By Lemma 4.2, for a fixed triple (a,t,i) there is at most one recorded action. Once i is enumerated into $S_{a,t}$, the corresponding stem and identifier can be recovered effectively from the historical record. Thus $S_{a,t}$ records the positive decisions made at address a during epoch t.

At a final epoch, no further injury by a proper ancestor slot is possible. Consequently, each exit eventually makes a permanent positive or negative decision about the corresponding open set $U_{|a|}$.

Lemma 4.5 (Decision at a final epoch)**.** *Fix an address a and an exit $i \in \mathbb{N}$. Put $t = \theta(a)$ and $r = p(a) \frown \eta_i$. Then*

$$i \in S_{a,t} \implies \mathcal{P} \cap [r] \subseteq U_{|a|}, \tag{4.8}$$

$$i \notin S_{a,t} \implies \mathcal{P} \cap [r] \cap U_{|a|} = \varnothing. \tag{4.9}$$

If $i \in S_{a,t}$ and its event has stem ρ and identifier u, then $g(a,i) = \rho$ and $\theta(a^\frown(i,b)) = u$ for $b < 2$. If $i \notin S_{a,t}$, then $g(a,i) = r$ and $\theta(a^\frown(i,b)) = t$ for $b < 2$.

Proof. Beginning of the final epoch. Let s_0 be the first stage at which a has its final epoch $t = \theta(a)$. By Lemma 4.2, for every $s \geq s_0$,

$$\theta_s(a) = t, \qquad p_s(a) = p(a), \qquad r_s(a,i) = r.$$

No proper ancestor slot of a acts after stage s_0.

At stage s_0, there is no current entry at any slot (a,j). Indeed, if $s_0 = 0$, the initial table is empty. If $s_0 > 0$, the action introducing this epoch is at a proper ancestor slot of a and deletes all entries at slots (a,j). Consequently, every entry subsequently present at (a,i) comes from an action recorded in $S_{a,t}$, and such an entry is never subsequently cancelled.

Positive decision. Suppose first that $i \in S_{a,t}$, and let (a,t,i,ρ,u) be its unique event. At stage u, the exit root is r, and the action chooses

$$[\rho] \subseteq \mathcal{P}_{u-1} \cap [r] \cap U_{|a|,u-1}.$$

By (3.9) and monotonicity,

$$\mathcal{P} \cap [r] \subseteq [\rho] \subseteq U_{|a|}.$$

This proves (4.8). The record $D_s(a,i) = (\rho,u)$ remains present for all $s \geq u$. The stem and epoch recursions therefore give

$$g(a,i) = \rho, \qquad \theta(a^\frown(i,b)) = u \quad (b < 2).$$

Negative decision. The negative case requires an argument: the absence of an action during the final epoch must be shown to imply that $\mathcal{P} \cap [r]$ is disjoint from $U_{|a|}$.

Suppose that $i \notin S_{a,t}$. There is no current entry at (a,i) at any stage $s \geq s_0$. Hence the default clauses of the recursions give

$$g_s(a,i) = r, \qquad \theta_s(a^\frown(i,b)) = t \quad (s \geq s_0,\ b < 2).$$

This proves the remaining assertions about the final stem and the child epochs.

To prove (4.9), suppose for a contradiction that

$$B \in \mathcal{P} \cap [r] \cap U_{|a|}.$$

Since the approximations to $U_{|a|}$ are increasing, there is $s_1 \geq s_0$ such that $B \in U_{|a|,s}$ for every $s \geq s_1$. Also $B \in \mathcal{P}_s$ for every s. Thus, for every $s \geq s_1$,

$$B \in \mathcal{P}_s \cap [r_s(a,i)] \cap U_{|a|,s}.$$

The test (3.7) is therefore positive at every visit to (a,i) after this point. The slot has no current entry, and the fair schedule visits it infinitely often. At such a visit the construction must make it act, recording an event with parameters (a,t,i) and hence enumerating i into $S_{a,t}$. This contradiction proves (4.9). □

5. Generalized lowness and specialness

The terms *stopped* and *complete* refer to the parsing in Lemma 4.3. In the decoding procedure below, n counts decoding steps, and p_n, t_n abbreviate the final values $p(a_n), \theta(a_n)$. They should not be confused with stage-n approximations from Section 3. We use the same local notation for the coding procedure in Section 6.

Lemma 5.1 (Decoding a complete path)**.** *For a complete $B \in \mathcal{P}$, the oracle $B \oplus 0'$ computes its sequence of addresses, final roots, final epochs, exit indices, and coding bits.*

Proof. We describe a single oracle procedure for all complete paths, with the computable tree T and the computable construction fixed throughout.

At step n, the procedure has a triple (a_n, p_n, t_n) such that a_n is the address at level n in the parsing of B,

$$p_n = p(a_n), \qquad t_n = \theta(a_n), \qquad p_n \prec B.$$

Initially this holds with

$$(a_0, p_0, t_0) = (\langle\rangle, \langle\rangle, 0).$$

Suppose the current triple is known. The tail of B after p_n does not belong to $\mathcal{Q} = [T]$, since B is complete. Using B and the computable tree T, find the shortest prefix of this tail which lies outside T. This is an exit, hence a member of the set E defined in Section 2. Recall that we fixed there a computable enumeration without repetitions

$$E = \{\eta_i : i \in \mathbb{N}\}.$$

Search this enumeration to find the unique index i_n.

Since the family $(S_{a,t})_{a,t}$ is uniformly c.e., the oracle $0'$ decides whether $i_n \in S_{a_n,t_n}$, uniformly in the finite parameters a_n, t_n, i_n. For a positive answer, search the historical record for the unique event

$$(a_n, t_n, i_n, \rho_n, u_n).$$

This search terminates. For a negative answer, put

$$\rho_n = p_n \frown \eta_{i_n}, \qquad u_n = t_n.$$

By Lemma 4.5, in either case

$$\rho_n = g(a_n, i_n), \qquad \theta(a_n \frown (i_n, b)) = u_n \quad (b < 2).$$

By (4.6), applied to the exit i_n, B extends the final stem ρ_n. Read

$$b_n = B(|\rho_n|)$$

and define

$$a_{n+1} = a_n \frown (i_n, b_n), \qquad p_{n+1} = \rho_n \frown b_n, \qquad t_{n+1} = u_n.$$

The final-root recursion gives $p_{n+1} = p(a_{n+1})$, and the preceding application of Lemma 4.5 gives $t_{n+1} = \theta(a_{n+1})$. The parsing rule identifies a_{n+1} as the next address of B, and $p_{n+1} \prec B$. Thus the invariant is preserved.

Every search in each step terminates for a complete B. Running finitely many steps computes the data at any specified level, uniformly from $B \oplus 0'$. The procedure never tests whether an epoch is final, nor is it given any stabilisation stage. Finality is preserved inductively from the initial triple by Lemma 4.5. □

Proposition 5.2. *Every member of $\mathcal{P}$ belongs to* GL_1.

Proof. Fix $B \in \mathcal{P}$. By Lemma 4.3, it is either stopped or complete. Suppose first that $B = p(a) \frown R$ is stopped, where $R \in \mathcal{Q}$. Since $p(a)$ is finite, adding or deleting this fixed prefix is computable, so $B \equiv_{\mathrm{T}} R$ and therefore $B' \equiv_{\mathrm{T}} R'$. Since $\mathcal{Q} \subseteq \mathrm{GL}_1$,

$$B' \equiv_{\mathrm{T}} R' \equiv_{\mathrm{T}} R \oplus 0' \equiv_{\mathrm{T}} B \oplus 0'.$$

Now suppose that B is complete. Given e, use Lemma 5.1 to compute its address a_{2e}, final epoch t_{2e}, and exit index i_{2e} at level $2e$, using $B \oplus 0'$. By the parsing rule,

$$B \in \mathcal{P} \cap [p(a_{2e}) \frown \eta_{i_{2e}}].$$

Therefore (2.3) and Lemma 4.5 give

$$e \in B' \iff B \in U_{2e} \iff i_{2e} \in S_{a_{2e},t_{2e}}.$$

Since the family $(S_{a,t})_{a,t}$ is uniformly c.e., the last condition is decidable using $0'$ once its finite parameters have been computed. Thus $B' \leq_{\mathrm{T}} B \oplus 0'$. Conversely, $B \leq_{\mathrm{T}} B'$ and $0' \leq_{\mathrm{T}} B'$ for every real B, so $B \oplus 0' \leq_{\mathrm{T}} B'$. Hence $B' \equiv_{\mathrm{T}} B \oplus 0'$ in this case as well. □

Remark 5.3. The conclusion is asserted pointwise; no single reduction uniform over all $B \in \mathcal{P}$ is claimed. On complete paths the decoder above is uniform. If $B = p(a) \frown R$ is stopped, the reduction may depend on the fixed finite prefix $p(a)$ and on a reduction witnessing $R' \leq_{\mathrm{T}} R \oplus 0'$ for the particular tail $R \in \mathcal{Q}$. No effective procedure distinguishing stopped paths from complete paths is required.

Proposition 5.4. *The class $\mathcal{P}$ is special.*

Proof. The class $\mathcal{P}$ is Π^0_1 by construction and is nonempty by Lemma 4.3. Every stopped member has the form $B = p(a) \frown R$, where $R \in \mathcal{Q}$. Removing this fixed finite prefix shows that $R \leq_{\mathrm{T}} B$, so B is noncomputable because $\mathcal{Q}$ is special.

Suppose for a contradiction that a complete member $B \in \mathcal{P}$ is computable. Choose e such that φ_e is its total binary characteristic function. By (2.4),

$$U_{2e+1} = 2^{\mathbb{N}} \setminus \{B\}.$$

Let a be the address of B at level $2e+1$, let $t = \theta(a)$, and let i be the corresponding exit index. Put $r = p(a) \frown \eta_i$. By the parsing rule,

$$B \in \mathcal{P} \cap [r].$$

If $i \in S_{a,t}$, (4.8) would give $B \in U_{2e+1}$, which is false. Thus $i \notin S_{a,t}$. By (4.9) and the displayed identity for U_{2e+1},

$$\mathcal{P} \cap [r] \subseteq \{B\}.$$

On the other hand, Lemma 4.5 gives $g(a,i) = r$. Since $r \prec B$, the next bit of B after r is $B(|r|)$. Let

$$b = 1 - B(|r|).$$

The final-root recursion gives

$$p(a \frown (i,b)) = g(a,i) \frown b = r \frown b.$$

By (4.5),

$$r \frown b \frown \mathcal{Q} = p(a \frown (i,b)) \frown \mathcal{Q} \subseteq \mathcal{P} \cap [r].$$

Since $\mathcal{Q}$ is nonempty, choose $R \in \mathcal{Q}$ and put

$$C = r \frown b \frown R.$$

Then $C \in \mathcal{P} \cap [r]$, while

$$C(|r|) = b \neq B(|r|),$$

so $C \neq B$. This contradicts $\mathcal{P} \cap [r] \subseteq \{B\}$. □

6. The Join Property and its local form

The goal of this section is to turn the permanent event record into the coding required by the Join Property. The key point is that, along a suitably chosen complete path, the coding can be decoded both from $B \oplus Z$ and from $B \oplus 0'$. We first establish an agreement lemma used to select the exits, then carry out the coding from an arbitrary final root. The availability of final roots inside every nonempty relatively basic open subclass yields the local form of the Join Property and completes the proof of the extension theorem.

For a real Z, write $\overline{Z} = \mathbb{N} \setminus Z$ for its complement; as characteristic functions, $\overline{Z}(n) = 1 - Z(n)$.

Lemma 6.1 (Agreement with a c.e. set)**.** *Let Z be noncomputable. One can choose $\widetilde{Z} \in \{Z, \overline{Z}\}$ so that $\overline{\widetilde{Z}}$ is not c.e. For every c.e. set S, there is an i satisfying*

$$i \in \widetilde{Z} \iff i \in S. \tag{6.1}$$

Given an index of S, such an i can be found using $\widetilde{Z} \oplus 0'$.

Proof. Since Z is noncomputable, Z and $\overline{Z}$ cannot both be c.e. If $\overline{Z}$ is not c.e., take $\widetilde{Z} = Z$; otherwise $\overline{Z}$ is c.e., so Z is not c.e., and we take $\widetilde{Z} = \overline{Z}$. In either case $\overline{\widetilde{Z}}$ is not c.e.

If (6.1) failed for every i, then $S = \overline{\widetilde{Z}}$, contradicting the fact that $\overline{\widetilde{Z}}$ is not c.e.

Given an index for S, the oracle $0'$ decides membership in S uniformly. Using also $\widetilde{Z}$, search for the least i on which S and $\widetilde{Z}$ agree. The preceding argument guarantees that this search terminates. □

The choice of $\widetilde{Z}$ is made once and for all and need not be uniform in Z. Once this choice is fixed, the search for an agreement is uniform in an index of S.

Proposition 6.2 (Coding from a final root)**.** *Fix an address a_* and reals A, Z with $A \geq_{\mathrm{T}} Z \oplus 0'$ and $Z >_{\mathrm{T}} 0$. There is a complete path $B \in \mathcal{P} \cap [p(a_*)]$ satisfying* (1.1)*.*

Proof. Fix $\widetilde{Z}$ as in Lemma 6.1. This orientation need not be chosen uniformly from Z; it is one fixed binary choice, and $\widetilde{Z} \equiv_{\mathrm{T}} Z$. Fix also the finite parameters

$$a_*, \qquad p(a_*), \qquad \theta(a_*),$$

together with an index for a Turing reduction witnessing $Z \oplus 0' \leq_{\mathrm{T}} A$. These finite parameters and this index may be built into the relevant oracle programs. The initial root and its epoch are treated here as fixed parameters; their $0'$-uniform recovery is discussed in Section 8.

Construction of the path. Using A, we construct triples (a_n, p_n, t_n) satisfying

$$a_* \preceq a_n, \qquad |a_n| = |a_*| + n, \qquad p_n = p(a_n), \qquad t_n = \theta(a_n).$$

Start with

$$(a_0, p_0, t_0) = (a_*, p(a_*), \theta(a_*)).$$

Given the current triple, choose the least i_n such that

$$i_n \in \widetilde{Z} \iff i_n \in S_{a_n, t_n}. \tag{6.2}$$

Lemma 6.1 guarantees existence. Since $\widetilde{Z} \equiv_{\mathrm{T}} Z$, we have $\widetilde{Z} \oplus 0' \leq_{\mathrm{T}} A$. As the family $(S_{a,t})_{a,t}$ is uniformly c.e., the least such i_n can therefore be found using A.

If $i_n \in S_{a_n,t_n}$, search the historical record for the unique event $(a_n, t_n, i_n, \rho_n, u_n)$. Otherwise put

$$\rho_n = p_n \frown \eta_{i_n}, \qquad u_n = t_n.$$

By Lemma 4.5, in either case

$$\rho_n = g(a_n, i_n), \qquad \theta(a_n \frown (i_n, b)) = u_n \quad (b < 2).$$

Set

$$a_{n+1} = a_n \frown (i_n, A(n)), \qquad p_{n+1} = \rho_n \frown A(n), \qquad t_{n+1} = u_n. \tag{6.3}$$

The final-root recursion gives $p_{n+1} = p(a_{n+1})$, and the preceding application of Lemma 4.5 gives $t_{n+1} = \theta(a_{n+1})$. Thus the induction invariant is preserved.

Moreover,

$$p_n \frown \eta_{i_n} \preceq \rho_n \prec p_{n+1}.$$

Since every exit is nonempty, the roots are nested and their lengths increase by at least two at each step. Let

$$B = \bigcup_n p_n.$$

Adjoining the proper prefixes of a_*, in increasing order of length, to the sequence $(a_n)_{n \in \mathbb{N}}$ gives an infinite address chain starting at $\langle\rangle$.

The final roots of these additional addresses are prefixes of $p(a_*) = p_0$, so their inclusion does not change the union. Lemma 4.3 therefore shows that B is complete and

$$B \in \mathcal{P} \cap [p(a_*)].$$

The reduction $B \leq_{\mathrm{T}} A$. With the fixed auxiliary data specified above, the sequence $(p_n)_{n \in \mathbb{N}}$ is computable from A.

To compute $B(k)$, construct roots until $|p_n| > k$ and return $p_n(k)$. This terminates because the lengths are unbounded. Hence

$$B \leq_{\mathrm{T}} A. \tag{6.4}$$

By construction, for every n,

$$B(|\rho_n|) = A(n).$$

Decoding from $B \oplus Z$. Use the fixed orientation to answer questions about $\widetilde{Z}$, and start with $(a_*, p(a_*), \theta(a_*))$. Inductively suppose the triple (a_n, p_n, t_n) used in the coding has been recovered. Using B and the computable tree T, find the shortest prefix of the tail after p_n

which lies outside T. This is the unique exit η_{i_n}, and its index i_n is recovered from the fixed computable enumeration of E. Thus the decoder recovers from B the exit selected during the coding; it does not repeat the least-agreement search. If $i_n \in \widetilde{Z}$, search the historical record for the event with parameters (a_n, t_n, i_n) and retrieve (ρ_n, u_n). This search terminates because (6.2) gives $i_n \in S_{a_n,t_n}$, so the corresponding event exists.

If $i_n \notin \widetilde{Z}$, the same equivalence guarantees that no such event exists, so use

$$\rho_n = p_n \frown \eta_{i_n}, \qquad u_n = t_n.$$

In either case these are exactly the data used in the coding. Read $b_n = B(|\rho_n|) = A(n)$ and update the triple to

$$(a_n \frown (i_n, b_n), \rho_n \frown b_n, u_n).$$

By (6.3), this is the next coding triple. Thus the induction continues and $A \leq_{\mathrm{T}} B \oplus Z$.

Decoding from $B \oplus 0'$. Start with the same fixed initial triple and recover the exit i_n from B exactly as above.

Instead of querying $\widetilde{Z}$, use $0'$ to decide $i_n \in S_{a_n,t_n}$. The positive or negative answer recovers the same ρ_n, u_n as above, after which the coding bit and the next triple are recovered in the same way. This decoder does not use Z. Consequently $A \leq_{\mathrm{T}} B \oplus 0'$. Together with $B \leq_{\mathrm{T}} A$ and $Z, 0' \leq_{\mathrm{T}} A$, the two decoding reductions give

$$A \equiv_{\mathrm{T}} B \oplus Z \equiv_{\mathrm{T}} B \oplus 0'.$$

By Proposition 5.2,

$$B' \equiv_{\mathrm{T}} B \oplus 0',$$

and hence all the equivalences in (1.1) hold. The agreement (6.2) is required only along the path selected for the given A, Z, not on every member of $\mathcal{P}$. □

Lemma 6.3 (Final roots are locally available)**.** *If $\mathcal{P} \cap [\sigma] \neq \varnothing$, some final root $p(a)$ extends σ.*

Proof. Choose $B \in \mathcal{P} \cap [\sigma]$.

If B is complete, let $(a_n)_{n\in\mathbb{N}}$ be its address chain. The roots $p(a_n)$ are prefixes of B and have unbounded lengths. Choose n with $|p(a_n)| \geq |\sigma|$. Since σ and $p(a_n)$ are both prefixes of B, we have $\sigma \preceq p(a_n)$.

Suppose instead that

$$B = p(a_0) \frown R, \qquad R \in \mathcal{Q}.$$

If $\sigma \preceq p(a_0)$, take $a = a_0$. Otherwise, since σ and $p(a_0)$ are both prefixes of B and $\sigma \not\preceq p(a_0)$, we have $p(a_0) \prec \sigma$. Hence there is a nonempty finite string δ such that

$$\sigma = p(a_0) \frown \delta, \qquad \delta \prec R.$$

In particular, $R \in \mathcal{Q} \cap [\delta]$, so Lemma 2.1 gives an exit η_i extending δ. By the final stem and root identities,

$$\sigma = p(a_0) \frown \delta \preceq p(a_0) \frown \eta_i \preceq g(a_0, i) \prec p(a_0 \frown (i, 0)).$$

Thus $a = a_0 \frown (i, 0)$ is a suitable address. □

Completion of the proof of Theorem 1.3. The class $\mathcal{P}$ is Π^0_1 by construction, and $\mathcal{Q} \subseteq \mathcal{P}$ by Lemma 4.3. Propositions 5.2 and 5.4 show that $\mathcal{P} \subseteq \mathrm{GL}_1$ and that $\mathcal{P}$ is special.

Fix a finite binary string σ such that $\mathcal{P} \cap [\sigma] \neq \varnothing$. By Lemma 6.3, choose an address a with $\sigma \preceq p(a)$. Hence $[p(a)] \subseteq [\sigma]$.

Let A, Z be arbitrary reals satisfying $A \geq_{\mathrm{T}} Z \oplus 0'$ and $Z >_{\mathrm{T}} 0$. Apply Proposition 6.2 with initial address $a_* = a$. It provides a complete path

$$B \in \mathcal{P} \cap [p(a)] \subseteq \mathcal{P} \cap [\sigma]$$

such that

$$A \equiv_{\mathrm{T}} B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} B \oplus Z.$$

Thus $\mathcal{P} \cap [\sigma]$ has the Join Property. Since σ was arbitrary, this proves the local conclusion and completes the theorem. □

7. Separation and a nontrivial pseudojump

We now derive the separation between the Join Property and pseudojump inversion. The decisive observation is that no class contained in GL_1 can have the Pseudojump Inversion Property. This completes the proof of the main theorem. We then record a stronger form of the obstruction, showing that the failure already occurs for a fixed nontrivial pseudojump operator which strictly raises every input degree.

Proposition 7.1. *No class $\mathcal{K} \subseteq \mathrm{GL}_1$ has the Pseudojump Inversion Property.*

Proof. For every $B \in \mathrm{GL}_1$ we have $0' \not\leq_{\mathrm{T}} B$. Otherwise,

$$B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} B,$$

contradicting strictness of the jump.

Let $\mathcal{K} \subseteq \mathrm{GL}_1$, and suppose that $\mathcal{K}$ has the Pseudojump Inversion Property. Fix an index $e_\varnothing$ such that

$$W_{e_\varnothing}^X = \varnothing \qquad \text{for every } X.$$

Then $J_{e_\varnothing}(B) \equiv_{\mathrm{T}} B$ for every B. Applying (1.2) with $A = 0'$ and $e = e_\varnothing$ gives a member $B \in \mathcal{K}$ satisfying

$$0' \equiv_{\mathrm{T}} J_{e_\varnothing}(B) \equiv_{\mathrm{T}} B.$$

This contradicts $0' \not\leq_{\mathrm{T}} B$, since $B \in \mathrm{GL}_1$. □

Proof of Theorem 1.2. Take a special Π^0_1 class $\mathcal{Q} \subseteq \mathrm{GL}_1$, as provided by [3, Theorem 6.2, parts 2 and 4], and apply Theorem 1.3. The resulting $\mathcal{P}$ has the Join Property, including its asserted local form, but does not have the Pseudojump Inversion Property by Proposition 7.1. □

The use of the pseudojump operator $J_{e_\varnothing}$ above is sufficient, but it is not essential. The following observation isolates the generalized-lowness argument used in [3, proof of Theorem 6.4].

Proposition 7.2. *Let J be a pseudojump operator such that*

$$X <_{\mathrm{T}} J(X) \quad \textit{and} \quad J(X)' \equiv_{\mathrm{T}} X' \qquad \textit{for every } X.$$

Then $0' \not\leq_{\mathrm{T}} J(B)$ for every $B \in \mathrm{GL}_1$.

Proof. If $0' \leq_{\mathrm{T}} J(B)$, the definition of a pseudojump also gives $B \leq_{\mathrm{T}} J(B)$. Hence

$$J(B)' \equiv_{\mathrm{T}} B' \equiv_{\mathrm{T}} B \oplus 0' \leq_{\mathrm{T}} J(B),$$

contradicting strictness of the jump. □

Such an operator is used in [3, proof of Theorem 6.4]. Consequently, for the constructed class $\mathcal{P}$, pseudojump inversion already fails for this single fixed operator: taking $A = 0'$, there is no $B \in \mathcal{P}$ for which

$$0' \equiv_{\mathrm{T}} J(B) \equiv_{\mathrm{T}} B \oplus 0'.$$

Indeed, Proposition 7.2 rules out even the first equivalence. Thus the obstruction persists for a pseudojump operator which strictly raises every input degree. This observation is not needed for the main separation.

8. Concluding remarks and further questions

We have shown that the Join Property and the Pseudojump Inversion Property can be separated on a special Π^0_1 class. In fact, the separation can be obtained under the additional restriction that every member of the class is generalized low, while the Join Property holds locally on every nonempty relatively basic open subclass. The extension theorem shows that this is not an isolated construction: every special Π^0_1 class contained in GL_1 can be enlarged, without leaving GL_1, to a class with the local Join Property.

The role of generalized lowness suggests a first direction for further study. In the present argument, GL_1 serves both as a class that can be preserved by the construction and as an

obstruction to pseudojump inversion. It would be natural to ask for which other degree-theoretic classes $\mathcal{C}$ an analogous extension theorem is possible: given a special Π^0_1 class $\mathcal{Q} \subseteq \mathcal{C}$, when can one find a special Π^0_1 extension $\mathcal{P}$ still contained in $\mathcal{C}$ and having the local Join Property? One may also ask how far the generalized-lowness hypothesis can be weakened while still obtaining a separation between the two properties.

A second issue concerns uniformity. For the fixed computable construction, the functions $a \mapsto p(a)$ and $a \mapsto \theta(a)$ are uniformly computable in $0'$, since their stage approximations are computable and eventually constant at each input. Hence, given σ with $\mathcal{P} \cap [\sigma] \neq \varnothing$, one can use $0'$ to search for an address a such that $\sigma \preceq p(a)$; Lemma 6.3 guarantees that this search terminates. The coding proof nevertheless fixes auxiliary data: an orientation $\widetilde{Z} \in \{Z, \overline{Z}\}$ with $\widetilde{\overline{Z}}$ not c.e., an index witnessing $Z \oplus 0' \leq_{\mathrm{T}} A$, and the finite initial data used by the decoders. In particular, the decoder from $B \oplus Z$ is not given $0'$ as an additional oracle. It is therefore natural to investigate stronger uniform local formulations in which the permitted oracle inputs and reduction indices are specified explicitly, and to determine which auxiliary choices can be eliminated.

Finally, Appendix A shows that every jump-inversion fibre above $0'$ of a class with the Join Property is countably infinite. It is natural to investigate the order-theoretic structure of these fibres and to determine which further constraints follow from additional hypotheses on the class.

More generally, since pseudojump operators admit finite and transfinite iterations, it would be interesting to investigate analogous class-restricted inversion and join phenomena for iterated pseudojumps and for higher levels of the corresponding hierarchies.

Appendix A. Infinitely many degrees in each Join Property fibre

This appendix is independent of the construction. It records the additional fibre result without assuming that the Turing jump preserves finite joins.

Write $\deg_{\mathrm{T}}(B)$ for the Turing degree of B, that is, its equivalence class under $\equiv_{\mathrm{T}}$.

For an arbitrary class $\mathcal{R} \subseteq 2^{\mathbb{N}}$ and $A \geq_{\mathrm{T}} 0'$, put

$$\mathcal{F}_{\mathcal{R}}(A) = \{\deg_{\mathrm{T}}(B) : B \in \mathcal{R} \ \&\ B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} A\}.$$

Thus $\mathcal{F}_{\mathcal{R}}(A)$ is the generalized-low part of the ordinary jump-inversion fibre. Recall that a real G is 1-*generic relative to* H if, for every H-c.e. set of strings $W \subseteq 2^{<\mathbb{N}}$, some prefix $\sigma \prec G$ satisfies

$$\sigma \in W \quad \text{or} \quad \{\tau \in W : \sigma \preceq \tau\} = \varnothing.$$

These alternatives say that G *meets* or *avoids* W, respectively. A set of strings is *dense* if every finite binary string has an extension in it. Finally, a $\Sigma^0_1(H)$ question is an existential question over an H-computable predicate, with finite parameters allowed; such questions are decidable using H'.

Lemma A.1. *Suppose $m \geq 1$ and $B_i' \leq_{\mathrm{T}} A$ for each $i < m$. There is a noncomputable $G \leq_{\mathrm{T}} A$ which is 1-generic relative to each B_i. Moreover,*

$$(B_i \oplus G)' \equiv_{\mathrm{T}} B_i' \oplus G \qquad (i < m).$$

Proof. Construction of G. For each $i < m$, fix an index witnessing $B_i' \leq_{\mathrm{T}} A$. Since the family is finite, these indices can be built into a single oracle program. Also $B_i \leq_{\mathrm{T}} B_i' \leq_{\mathrm{T}} A$ for every $i < m$.

View $W_e^{B_i}$ as a set of finite binary strings under a fixed computable coding. Fix a computable enumeration without repetitions of all pairs (i, e) with $i < m$ and $e \in \mathbb{N}$. Using A, construct a nested sequence of strings $(\sigma_s)_{s \in \mathbb{N}}$, starting with the empty string.

At the step for (i, e), use B_i' to decide whether

$$\exists \tau \succeq \sigma_s \ (\tau \in W_e^{B_i}).$$

This is a $\Sigma^0_1(B_i)$ question, uniformly in the finite parameters. If the answer is positive, enumerate $W_e^{B_i}$ until such a τ appears. If the answer is negative, put $\tau = \sigma_s$. In either case set

$$\sigma_{s+1} = \tau \frown 0.$$

All these operations are computable in A.

Let $G = \bigcup_s \sigma_s$. The lengths increase at every step, so $G \leq_{\mathrm{T}} A$. If a requirement receives a positive answer, the chosen $\tau \in W_e^{B_i}$ remains a prefix of G. If it receives a negative answer, σ_s remains a prefix of G with no extension in $W_e^{B_i}$. Thus G is 1-generic relative to each B_i.

Noncomputability. For every $i < m$, this implies $G \not\leq_{\mathrm{T}} B_i$. Indeed, if $G \leq_{\mathrm{T}} B_i$, then

$$D_G = \{\sigma \in 2^{<\mathbb{N}} : \exists k < |\sigma|\ [\sigma(k) \neq G(k)]\}$$

would be a dense B_i-computable set of strings. No prefix of G belongs to D_G, and density prevents G from avoiding it, contrary to relative genericity. Since $m \geq 1$, fixing any $i < m$ gives $G \not\leq_{\mathrm{T}} B_i$, and hence G is noncomputable.

The jump of $B_i \oplus G$. Finally, fix $i < m$. For each e, let

$$C_{i,e} = \{\tau \in 2^{<\mathbb{N}} : \Phi_e^{B_i \oplus \tau}(e) \downarrow\}.$$

Here $\Phi_e^{B_i \oplus \tau}(e) \downarrow$ means that the computation converges using only bits of the finite second oracle below $|\tau|$. Consequently, this finite computation persists, with the same outcome, for every infinite extension of τ used as the second oracle. The set $C_{i,e}$ is upward closed and uniformly c.e. in B_i.

Since G is 1-generic relative to B_i, it either meets $C_{i,e}$ or has a prefix with no extension in $C_{i,e}$. Using $B_i' \oplus G$, examine successive prefixes σ of G. The oracle B_i' decides both whether $\sigma \in C_{i,e}$ and whether some extension of σ belongs to $C_{i,e}$. If $\sigma \in C_{i,e}$, answer that $e \in (B_i \oplus G)'$. If no extension of σ belongs to $C_{i,e}$, answer that $e \notin (B_i \oplus G)'$. By the preceding genericity alternative, one of these cases is eventually found.

The positive answer is correct because the finite computation witnessed by the prefix σ persists along G. The negative answer is correct because any convergence with oracle $B_i \oplus G$ would be witnessed by a sufficiently long prefix of G extending σ. Hence

$$(B_i \oplus G)' \leq_{\mathrm{T}} B_i' \oplus G.$$

Conversely, monotonicity of the jump gives $B_i' \leq_{\mathrm{T}} (B_i \oplus G)'$, and $G \leq_{\mathrm{T}} B_i \oplus G \leq_{\mathrm{T}} (B_i \oplus G)'$. Therefore

$$(B_i \oplus G)' \equiv_{\mathrm{T}} B_i' \oplus G.$$

□

Proposition A.2. *If $\mathcal{R}$ has the Join Property, then $\mathcal{F}_{\mathcal{R}}(A)$ is countably infinite for every $A \geq_{\mathrm{T}} 0'$.*

Proof. Fix $A \geq_{\mathrm{T}} 0'$. Since $0' >_{\mathrm{T}} 0$ and $0' \oplus 0' \equiv_{\mathrm{T}} 0' \leq_{\mathrm{T}} A$, the Join Property applied with $Z = 0'$ gives a real $C \in \mathcal{R}$ such that

$$C' \equiv_{\mathrm{T}} C \oplus 0' \equiv_{\mathrm{T}} A.$$

Thus $\mathcal{F}_{\mathcal{R}}(A)$ is nonempty.

If B represents a degree in this fibre, then $B \leq_{\mathrm{T}} B' \equiv_{\mathrm{T}} A$, and hence $B \leq_{\mathrm{T}} A$. Since only countably many reals are computable in a fixed oracle, $\mathcal{F}_{\mathcal{R}}(A)$ is at most countable.

Suppose for a contradiction that the fibre is finite. Choose representatives $B_0, \ldots, B_{m-1} \in \mathcal{R}$, with $m \geq 1$, for all its degrees, so that

$$B_i' \equiv_{\mathrm{T}} B_i \oplus 0' \equiv_{\mathrm{T}} A \qquad (i < m).$$

Apply Lemma A.1 to this finite family. It gives a noncomputable $G \leq_{\mathrm{T}} A$ such that

$$(B_i \oplus G)' \equiv_{\mathrm{T}} B_i' \oplus G \equiv_{\mathrm{T}} A \qquad (i < m),$$

where the last equivalence uses $B_i' \equiv_{\mathrm{T}} A$ and $G \leq_{\mathrm{T}} A$.

For each $i < m$, we have $A \not\leq_{\mathrm{T}} B_i \oplus G$. Indeed, otherwise

$$(B_i \oplus G)' \leq_{\mathrm{T}} A \leq_{\mathrm{T}} B_i \oplus G,$$

contradicting strictness of the jump.

Since G is noncomputable and $G \oplus 0' \leq_{\mathrm{T}} A$, the Join Property, now applied with $Z = G$, supplies a real $B \in \mathcal{R}$ satisfying

$$A \equiv_{\mathrm{T}} B' \equiv_{\mathrm{T}} B \oplus 0' \equiv_{\mathrm{T}} B \oplus G.$$

Thus $\deg_{\mathrm{T}}(B) \in \mathcal{F}_{\mathcal{R}}(A)$, so $B \equiv_{\mathrm{T}} B_j$ for some $j < m$. Consequently,

$$A \equiv_{\mathrm{T}} B \oplus G \equiv_{\mathrm{T}} B_j \oplus G,$$

contrary to $A \not\leq_{\mathrm{T}} B_j \oplus G$.

Hence the fibre is infinite, and therefore countably infinite. □

Consequently, if $\mathcal{R}$ has the Join Property and $A \geq_{\mathrm{T}} 0'$, the ordinary jump-inversion fibre $\{\deg_{\mathrm{T}}(B) : B \in \mathcal{R} \ \&\ B' \equiv_{\mathrm{T}} A\}$ is also countably infinite: it contains $\mathcal{F}_{\mathcal{R}}(A)$, and every one of its representatives is computable in A.

Remark A.3. The genericity construction uses the reductions $B_i' \leq_{\mathrm{T}} A$ separately. In particular, it does not assume that the jump of $B_0 \oplus \cdots \oplus B_{m-1}$ is computable in A.

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